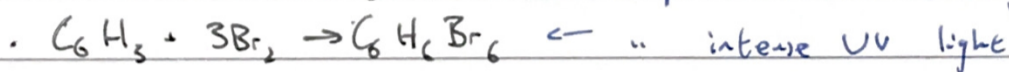
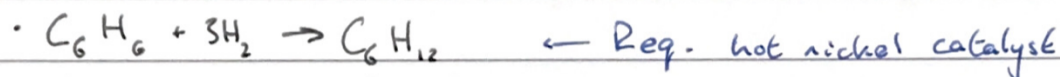


## Free-Radical Addition (Benzene)

→ Requires Nickel catalyst or UV light.

"Due to stability of benzene ring, most benzene reactions are sub. → Must have UV or catalyst to create highly reactive free radicals"



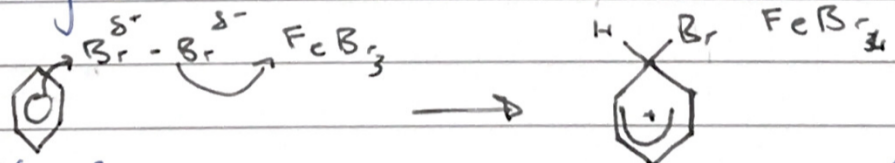
## Electrophilic substitution

→ Halogenation

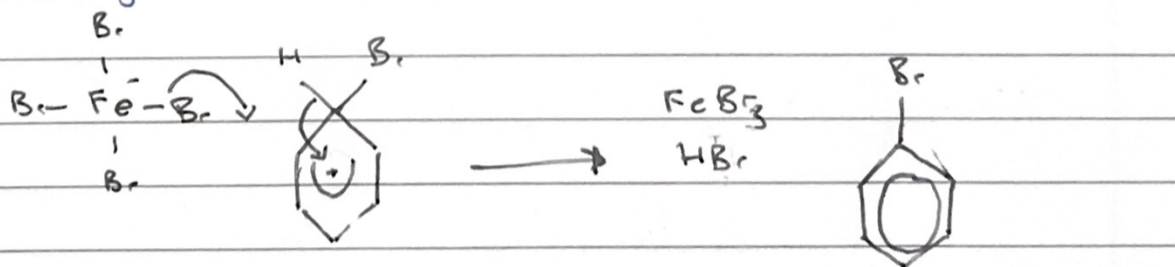
• Dry benzene reacts with liquid bromine in presence of iron

• Steaming HBr fumes given off.

Stage 1:



Stage 2:

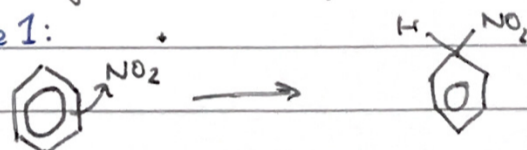


→ Nitration

• Benzene warmed with mixture of conc. nitric & sulphuric acids

• A nitro group ( $NO_2$ ) replaces one of the Hydrogen atoms.

Stage 1:



Stage 2:

